

A Post-Quantum Decentralized Threshold Authenticated Key Agreement Scheme for Securing Vehicular Ad-Hoc Networks

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Abstract—Authenticated key agreement (AKA) between vehicles and roadside units (RSUs) is a fundamental component for securing Vehicular ad-hoc networks (VANETs). However, most existing AKA schemes rely on conventional cryptographic primitives, such as discrete logarithm and large integer factorization, rendering them vulnerable to emerging quantum attacks. Although several post-quantum solutions based on lattice-hard problems have been proposed, their heavy dependence on a centralized trusted authority exposes them to single-point-of-failure risks. To address these issues, we propose a decentralized authenticated key agreement scheme grounded in the security assumptions of the Ring Learning With Errors (RLWE) and Inhomogeneous Small Integer Solution (ISIS) problems. The proposed scheme achieves strong resistance against quantum adversaries while preserving the privacy of communicating entities. Furthermore, a threshold-based voting mechanism is employed to mitigate the inherent key exposure problem within network infrastructure. By distributing authentication authority across multiple edge server nodes, the scheme significantly enhances system robustness against single-node failures and collusion attacks. Comprehensive security analysis and performance evaluations demonstrate that the proposed scheme satisfies diverse security requirements with acceptable computational and communication overhead, thereby confirming its practicality for deployment in real-world VANETs environments.

Index Terms—Vehicular ad-hoc networks, authenticated key agreement, post-quantum security, threshold, consortium blockchain.

I. INTRODUCTION

THE vehicular ad-hoc networks (VANETs) represent a special type of mobile ad-hoc network that uses wireless communication technologies to dynamically organize communication among moving nodes, thereby exhibiting high flexibility and self-organization. VANETs enable critical functions such as real-time safety alerts, autonomous driving, cooperative

driving, and road condition monitoring, serving as an effective solution to traffic congestion [1]. However, due to node mobility, rapidly changing network topology, and open wireless channels, VANETs face significant challenges in communication security, data reliability, and privacy protection.

In vehicle-to-infrastructure (V2I) communication, RSUs provide services to vehicles via open wireless channels, which are vulnerable to eavesdropping, forgery, and replay attacks [2], [3]. To counter these threats, the AKA protocol ensures mutual authentication and negotiates a session key between vehicles and infrastructure. This guarantees communication authenticity and provides confidentiality, integrity, and freshness for subsequent data transmission, making it essential for securing V2I channels.

Most existing AKA schemes in VANETs rely on a fully centralized Trusted Authority (TA) [4], [5], [6], [7], [8]. Although such schemes exhibit advantages including high computational efficiency and relative simplicity of deployment in specific scenarios, their strong reliance on a single TA often introduces risks such as single points of failure and vulnerabilities to Denial-of-Service (DoS) attacks. Consequently, they struggle to meet the security and privacy requirements inherent to the dynamic and open environment of vehicular networks. To address these issues, recent research has shifted towards decentralized security architectures. In these approaches, edge computing nodes share the authentication load, while blockchain technology enables distributed identity management and immutable record keeping [9], [10]. Furthermore, distributed identity authentication protocols based on blockchain have been proposed, such as those in [11], [12], [13], [14], [15]. By distributing trust across multiple participants, these approaches effectively enhance the system's robustness, scalability, and privacy protection capabilities.

Under the framework of traditional cryptography, the AKA protocols widely employed in VANETs are predominantly constructed based on classical cryptographic primitives such as Diffie-Hellman or Elliptic Curve Diffie-Hellman [16]. With the advancement of quantum computing technology, particularly the proposal and progressive realization of Shor's algorithm [17], these cryptographic mechanisms founded on the difficulty of integer factorization or discrete logarithm problems face severe threats [18], [19]. Quantum computers can solve the aforementioned classical hard problems in polynomial time, potentially rendering the existing VANETs security infrastructure completely vulnerable under quantum attacks. This could lead to

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serious consequences including identity spoofing, key leakage, and communication interception. Therefore, researchers have been actively working on designing AKA protocols capable of resisting quantum attacks.

Against this backdrop, AKA schemes based on post-quantum cryptographic primitives such as lattice-based cryptography have been successively proposed [20], [21], [22], [23], aiming to maintain secure communication in the face of quantum attacks. While early lattice-based protocols theoretically possess quantum-resistant properties, they often fail to balance communication security with VANET-specific requirements like conditional privacy protection in practical deployments. Subsequent research has further introduced post-quantum AKA protocols supporting privacy protection [24], [25], however, their overall security still relies on a single, centralized TA. This makes the system vulnerable to single points of failure and attacks targeting the centralized architecture, thus struggling to meet the distributed and highly dynamic application scenarios of VANETs. Consequently, constructing a decentralized, quantum-resistant, and practically deployable AKA scheme for vehicular networks has become a crucial and urgent issue to address.

To address the aforementioned challenges, we propose a decentralized, post-quantum secure AKA scheme based on the Ring Learning With Errors (RLWE) and the Inhomogeneous Small Integer Solution (ISIS) security assumptions for a consortium blockchain-driven VANETs. The core idea of the scheme is to construct a distributed AKA framework within a semi-trusted environment, enabling vehicles, RSUs, and Edge Servers (ES) to achieve secure mutual authentication and session key establishment in an open setting. Specifically, our scheme leverages the hardness of the ISIS problem to construct unforgeable distributed authentication tokens for decentralized identity verification. Concurrently, it utilizes the hardness of the RLWE problem to design an efficient key exchange mechanism for secure session key agreement. A cryptographic architecture based on both ISIS and RLWE is employed to accomplish mutual authentication and negotiate a session key without requiring a fully trusted environment. To completely eliminate reliance on a single trusted authority, we introduce a threshold-based distributed voting mechanism, dispersing the system's authentication authority among multiple semi-trusted ES nodes. These ES nodes collaborate as equal and independent entities via the consortium blockchain, forming a distributed identity verification network. The credibility of AKA messages between vehicles and RSUs depends on whether the number of authentication tokens issued for them by ES nodes reaches a preset threshold, thereby achieving decentralized trust assurance. The contributions of our proposed scheme are summarized as follows:

- 1) To achieve post-quantum security for V2I communication, we propose a decentralized, post-quantum secure AKA protocol based on the security assumptions of the ISIS problem and the RLWE problem in lattice-based cryptography. This design not only clarifies the protocol structure, but also reduces its overall security to two distinct hard lattice problems, thereby enhancing robustness while leveraging the efficiency and functionality advantages of each primitive. Unlike existing lattice-based schemes, this

protocol leverages a consortium blockchain and semi-trusted edge nodes as a distributed trust infrastructure. It enables authentication and key agreement between vehicles and RSUs, while preserving communication integrity and privacy. To the best of our knowledge, this is the first post-quantum distributed AKA for VANETs.

- 2) Our scheme designs a threshold-based distributed voting mechanism, dispersing the system secret among multiple ES nodes, thereby completely eliminating reliance on a centralized entity or a primary key. This mechanism operates on the consortium blockchain, ensuring that the system can function securely even if some ES nodes are compromised or offline, demonstrating resilience against single points of failure and collusion attacks. Furthermore, we employ smart contracts for public key and token management. While guaranteeing post-quantum security, this approach significantly reduces the computational and communication overhead for vehicles, making the scheme more suitable for the resource-constrained VANETs environment.
- 3) We conducted a security verification and analysis of the proposed scheme. Security proof show that the protocol not only ensures message authentication and integrity, non-repudiation, anonymity, unforgeability, and anti-eavesdropping, but also effectively resists most known attacks. Furthermore, compared with other similar schemes, our scheme has acceptable computational and communication overhead, thus demonstrating its potential usability.

II. RELATED WORK

Most existing VANETs AKA schemes are based on a centralized trusted architecture, which relies on a fully trusted center to handle the registration, authentication, revocation, and other functions of VANETs communication entities. However, its inherent centralized drawbacks have also driven the evolution of VANETs security architecture towards a more distributed and robust direction. We reviewed the AKA schemes of VANETs and summarized the technologies, advantages and limitations used by these schemes in Table I.

Zhu et al. [4] proposed an efficient novel pairing-free CLAS scheme and claimed that its construction could not only resist full key attacks, but also support any verifier in checking the validity of aggregated signatures. However, it was later pointed out by Yang et al. [5] that the scheme had security flaws involving four types of forgery attacks. Wang et al. [6] introduced a new fuzzy certificateless signature authentication scheme, which effectively protects the confidentiality of the real biometric features of users while achieving conditional privacy, but did not consider protection against key exposure. Liu et al. [7] proposed a lightweight anonymous authentication key agreement scheme, which accelerates authentication by pre-synchronizing vehicle authentication information to target RSUs through trajectory planning. Nevertheless, its reliance on a fully trusted authority for path planning and pseudonym management, and its dependence on a single TA make it vulnerable to DoS attacks. Li et al. [8] presented an efficient anonymous

TABLE I
COMPARATIVE SUMMARY OF RELATED WORKS

References	Techniques	Advantages	Limitations
[4], [6]	Hash function and certificateless signature	Supporting confidentiality and conditional privacy	Failing to support post-quantum security and failing to consider protection against key exposure
[7], [8]	Hash function, PUF, and ECC	Supporting confidentiality, anonymity, and formal analysis	Failing to support post-quantum security and require fully-trusted environment
[26]	Blockchain, PUF, and ECC	Supporting confidentiality, anonymity, and formal analysis	Lack of post-quantum security, conditional privacy, and voucher revocation mechanism
[27]	Blockchain, Hash function, and Bilinear Pairing	Supporting confidentiality, formal analysis, and cross-domain identity management	Failing to support post-quantum security and require fully-trusted environment
[29], [32]	Hash function, ECC, and Fog computing	Supporting confidentiality, anonymity, and batch verification	Failing to support post-quantum security
[20], [21], [22]	Hash function and RLWE-based key agreement	Supporting confidentiality and formal analysis	Lack of conditional privacy and require fully-trusted environment
[23]	Hash function, LWE-based key agreement, and ISIS-based authenticated	Supporting confidentiality, anonymity, and formal analysis	Lack of conditional privacy and lightweight deployment
[33], [34]	Hash function and RLWE-based key agreement	Supporting confidentiality and formal analysis	Failing to support tracing robustness and key escrow freeness
[24]	Hash function and RLWE-based key agreement	Supporting confidentiality, anonymity, and formal analysis	Lack of lightweight deployment and require fully-trusted environment
[35]	QKD, Hash function and CRYSTALS-Kyber KEM	Supporting confidentiality, anonymity, and formal analysis	Failing to support lightweight deployment, require fully-trusted environment, and relies on dedicated hardware
[36]	Blockchain, Hash function and ISIS-based authenticated	Supporting confidentiality, formal analysis, and cross-domain authentication	Depend on multiple authoritative institutions and require fully-trusted environment

authentication and session key agreement scheme based on elliptic curve cryptography and physical unclonable functions, enabling mutual authentication, anonymous communication, and lightweight session key negotiation between vehicles and RSUs, yet it still relies on a fully trusted authority for registration and revocation management.

Although the aforementioned schemes address key issues in authentication key agreement to varying degrees, most of them are essentially centralized. To tackle the inherent performance and security problems of centralized VANETs communication architectures, Badshah et al. [26] proposed a blockchain-based anonymous key exchange scheme AAKE-BIVT, achieving multi-level secure session key agreement among vehicles, between vehicles and RSUs, and between RSUs and cloud servers. However, this scheme does not support conditional privacy protection and credential revocation mechanisms, limiting its applicability in dynamic vehicular environments. Recently, Li et al. [27] introduced a blockchain-based revocable authentication scheme for vehicular networks that supports cross-domain identity management, enabling secure access between domain vehicles and anonymous communication between different administrative technologies and identity systems. Meanwhile, several blockchain-based authentication protocols for VANETs have been designed in the existing literature, such as [11], [12], [13], [14], [15]. Among them, Dwivedi et al. [15] utilized blockchain for vehicle pre-authentication and used ECC and one-way hash functions to verify individual messages, improving the authenticity of the message, but potentially increasing the latency for high-frequency data exchange. In addition, several fog computing-based authentication schemes for VANETs have been proposed [28], [29], [30], [31]. Al-Mekhlafi et al. [28] proposed a certificateless aggregate authentication scheme based on fog computing, which effectively reduces computational and

communication overhead. Subsequently, Almazroi et al. [32] integrated blockchain with fog computing to design an authentication scheme for vehicular blockchain networks, enhancing the security and traceability of vehicle communications by combining smart contracts and channel feature extraction techniques.

However, both centralized and distributed protocols remain susceptible to quantum attacks, and the rapid progress of quantum technologies renders conventional public-key cryptography increasingly insecure, necessitating post-quantum solutions [37], [38].

Nowadays, many post-quantum protocols [20], [21], [22] are proposed based on hard lattice problems. The RLWE problem, as a fundamental variant of lattice problems, has garnered significant attention due to its flexibility, computational efficiency, small key sizes, and high security. Dabra et al. [20] specifically designed a lattice-based anonymous password-authenticated key exchange protocol for mobile devices. However, it still cannot resist denial-of-service attacks. Wei et al. [23] introduced a lattice-based certificateless anonymous AKA scheme to secure server-to-server communication in distributed IoT systems. This scheme combines LWE and ISIS security assumptions, resulting in lower computational efficiency compared to some Ring-LWE-based schemes. Furthermore, these schemes do not satisfy the conditional preservation of privacy.

Bagchi et al. [39] proposed a blockchain-based post-quantum lattice aggregate signature scheme for drone IoT applications, leveraging the hardness of the RLWE problem to achieve efficient signature aggregation and verification. However, it relies on a trusted key generation center and might face performance bottlenecks on drone nodes. Khalid et al. [33] proposed a novel post-quantum secure protocol named RAVEN, which focuses on end-to-end anonymous encryption and efficient mutual authentication for smart vehicles. Xu et al. [34] introduced

a certificateless aggregate signature scheme based on lattices that not only enhances security but also provides privacy protection features. However, these works focus on authentication rather than authenticated key agreement, and they do not consider the robustness of tracing and freedom from key escrow. Liu et al. [24] achieved quantum-safe anonymous authentication with robust traceability in VANETs by combining the RLWE problem, fuzzy expiration time, and a double-commitment mechanism, but the centralized management of certificates and associated overhead costs limit its applicability in resource-constrained scenarios. Cui et al. [25] proposed an RLWE-based post-quantum secure three-party AKA scheme suitable for digital twin environments in the Internet of Vehicles. This scheme achieves mutual authenticated key agreement among the vehicle, digital twin, and trusted authority in a single round of interaction, possessing resistance against quantum attacks and key reuse attacks. Zhong et al. [40] and Kumar et al. [41] respectively proposed lightweight authentication and key exchange schemes based on RLWE, suitable for resource-constrained IoT environments. Almazroi et al. [42] proposed a lattice-based conditional privacy-preserving authentication scheme, where a fog server generates secret keys and delivers them to registered vehicles via 5G-BS, while the trusted authority stores the primary secret key for each fog server. Parameswarath et al. [35] proposed a quantum-secure mutual authentication protocol that combines QKD with the post-quantum cryptography Kyber to achieve quantum security for platoon communication. This scheme balances lightweight design and privacy protection, but is based on specialized hardware. Yang et al. [36] proposed a lattice-based online/offline cross-domain authentication scheme for V2G networks, which reduces computational overhead by introducing online/offline signature technology and leverages blockchain for information storage; however, the core functions of the protocol still rely on multiple trusted authorities.

The aforementioned schemes all rely on a centralized security architecture anchored by a trusted third party. Although such an architecture facilitates strong centralized control and streamlined security management, it inherently introduces a single point of failure. Should the central server fail or become compromised, the entire VANETs security system would be paralyzed, hindering new vehicle registrations and disrupting ongoing authentications. Moreover, because the security of these schemes rests entirely on the assumption of an absolutely trustworthy authority, any malicious insider or successful compromise of the TA would lead to catastrophic consequences. In contrast, our scheme eliminates the reliance on any fully trusted centralized entity by utilizing semi-trusted edge servers as the sole trust infrastructure for authentication services. A threshold-based voting mechanism further removes dependence on a third-party authority, enabling decentralized trust and inherent resilience against single points of failure within a consortium blockchain framework.

III. PRELIMINARIES AND BACKGROUND

In this section, we review the fundamentals of lattice-based cryptography, including topics such as lattices, secret sharing

schemes, blockchain technology and smart contract, system model, and threat model.

A. Lattice Cryptography

Definition 1 (RLWE Problem): Let $n \geq 2$ be even, $q > 2$ prime, and $\mathbb{Z}_q[x]$ the polynomial ring over $\mathbb{Z}_q$. Define $R_q = \mathbb{Z}_q[x]/(x^n + 1)$. Let χ be a discrete Gaussian distribution over R_q with standard deviation β , which directly controls the magnitude of the random noise introduced by all entities. For parameters m, q, a , uniformly sample $a = (a_1, \dots, a_m)^\top \in R_q^m$ and $s, e \leftarrow \chi$. The problem is to distinguish between $(a, b = as + e)$ and uniform (a, r) [43].

In RLWE-based key agreement, since both communicating parties introduce their own random noises, a reconciliation mechanism is required to ensure that both parties can extract exactly the same session key from these noisy values. We employ a Cha algorithm that includes two mutually offset decision intervals with signal functions $\kappa_0(x)$ and $\kappa_1(x)$.

Definition 2 (Cha Function): Define two "signal" functions $\kappa_0(x)$ and $\kappa_1(x)$ from $\mathbb{Z}_q$ to $\{0, 1\}$ as follows:

$$\kappa_0(x) = \begin{cases} 0, & x \in [-\lfloor \frac{q}{4} \rfloor, \lfloor \frac{q}{4} \rfloor]; \\ 1, & \text{otherwise.} \end{cases}$$

$$\kappa_1(x) = \begin{cases} 0, & x \in [-\lfloor \frac{q}{4} \rfloor + 1, \lfloor \frac{q}{4} \rfloor + 1]; \\ 1, & \text{otherwise.} \end{cases}$$

We define an algorithm Cha as: for any $x \in \mathbb{Z}_q$, $\text{Cha}(x) = \kappa_b(x)$, where $b \xleftarrow{R} \{0, 1\}$.

Definition 3 (Mod Function): The mod function is defined as: $\text{Mod}_2(x, w) = (x + w \cdot \frac{q-1}{2} \bmod q) \bmod 2 \in \{0, 1\}$, where $w = \text{Cha}(x) \in \{0, 1\}$.

Lemma 1: [44]: For any ring elements $a, b \in R_q$, we have $\|ab\| \leq \sqrt{n}\|a\| \cdot \|b\|$ and $\|ab\|_\infty \leq \sqrt{n}\|a\|_\infty \|b\|_\infty$.

Lemma 2: [45]: For any positive real number $\beta \geq \omega(\sqrt{\log n})$, there holds $\Pr_{x \leftarrow \chi_\beta^n}[\|x\| > \beta\sqrt{n}] \leq 2^{-n+1}$.

Lemma 3: [46]: Given a prime $q > 2$, assume that $a, b \in R_q$ and $a = b + 2e$, where $e \in \chi_\beta^m$ and $\|e\| < \frac{q}{8} - 1$. Then, $\text{Mod}_2(a, \text{Cha}(a)) = \text{Mod}_2(b, \text{Cha}(a))$ holds.

Definition 4 (ISIS Problem): Let q be a positive integer, β a real number, and $A \in \mathbb{Z}_q^{n \times m}$ a random matrix. The Small Integer Solution (SIS) problem is to find a non-zero vector $e \in \mathbb{Z}_q^m$ that satisfies $Ae = 0$ and $\|e\| \leq \beta$. In particular, given a non-zero vector $u \in \mathbb{Z}_q^n$, the Inhomogeneous Small Integer Solution (ISIS) problem is to find $e \in \mathbb{Z}_q^m$ that satisfies $Ae = u \bmod q$ and $\|e\| \leq \beta$.

Definition 5 (Trapdoor on the ISIS): The trapdoor function in the ISIS assumption lets authorized entities compute secret values from public information using one-way functions.

- **TrapGen(p, k, ϑ)** [47]: Input an integer $p, \vartheta = \text{poly}(p)$, $k \geq 5p \log \vartheta$. This trapdoor generation algorithm outputs a random matrix $A \in \mathbb{Z}_\vartheta^{p \times k}$ and a trapdoor matrix $T \in \mathbb{Z}_\vartheta^{k \times k}$, where A is uniformly distributed with rank p , and T is a basis for the lattice $\Lambda_\vartheta^\perp(A)$, satisfying $AT = 0 \bmod \vartheta$, and $\|T\| \leq O(\sqrt{p \log \vartheta})$, where $O()$ is the time complexity function.

- *SamplePre*(A, T, h, σ) [48]: Input a matrix $A \in \mathbb{Z}_\theta^{p \times k}$, a matrix $T \in \mathbb{Z}_\theta^{k \times k}$, a Gaussian parameter $\sigma \leq \|T\| \cdot \omega(\sqrt{\log p})$, and a random vector $h \in \mathbb{Z}_\theta^p$, where $\omega(\cdot)$ is the space complexity function, and T is a basis for the lattice $\Lambda_\theta^\perp(A)$. This preimage sampling algorithm outputs a vector $e \in \mathbb{Z}_\theta^k$, with $\|e\| \leq \sigma\sqrt{k}$, obeying the discrete Gaussian distribution $D_{\Lambda_\theta^\perp(A), \sigma}$, and satisfying $Ae = h \bmod \vartheta$.
- *SampleRwithBasis*(A, T, U, σ) [49]: Input a matrix $A \in \mathbb{Z}_\theta^{p \times k}$, a matrix $T \in \mathbb{Z}_\theta^{k \times k}$, a Gaussian parameter $\sigma \leq \|T\| \cdot \omega(\sqrt{\log p})$, and a matrix $U \in \mathbb{Z}_\theta^{p \times k}$, where T is a basis for the lattice $\Lambda_\theta^\perp(A)$. This algorithm outputs a matrix $C \in \mathbb{Z}_\theta^{k \times k}$, where each column c_i satisfies $\|c_i\| \leq \sigma\sqrt{k} = O(p \log k) \cdot \omega(\sqrt{\log k})$, and the matrix C satisfies $AC = U \bmod \vartheta$.

B. Secret Sharing Schemes

Secret sharing, particularly (t, n) threshold secret sharing, is a method that divides a secret S into n shares $s_1, \dots, s_n$ so that any t or more shares can reconstruct S , while any set of fewer than t shares reveals no information about the secret. The scheme is composed of two algorithms: *Secret Sharing*(S, t, n) to split S , and *Secret Reconstruction*($s_1, \dots, s_t, t$) to recover S from at least t shares. This structure formally guarantees that for any $k < t$, $\Pr(\text{Recovery}(s_1, \dots, s_k) \rightarrow S) \leq \varepsilon$, thus satisfying the required security condition [50].

C. Blockchain Technology and Smart Contract

Blockchain is a distributed ledger allowing secure and transparent transactions among multiple parties. Its decentralized nature makes transactions more secure and transparent, eliminating the need for third-party interference. A smart contract is a self-executing program stored on a blockchain that automatically enforces agreements when predefined conditions are met. In this protocol, the blockchain primarily serves as a foundation for decentralized trust and state recording, executing only lightweight, deterministic logic such as counting, querying, and triggering state changes.

D. System Model

In our proposed scheme, the system model mainly includes four types of entities including the system initiator, edge server (ES), roadside unit (RSU), and vehicle, which are defined as follows and shown in Fig. 1.

- *System Initiator*: The system initiator is mainly responsible for system initialization and infrastructure management. This includes generating security parameters, selecting edge servers for vehicle authentication, and managing and maintaining the blockchain network. After the system initialization is completed, the system initiator no longer participates in the subsequent authentication and key negotiation processes.
- *ES*: In this scheme, the ES is positioned as a semi-trusted network entity. It is primarily responsible for performing vehicle identity authentication and determining the legitimacy of a specific vehicle based on its credit value. In

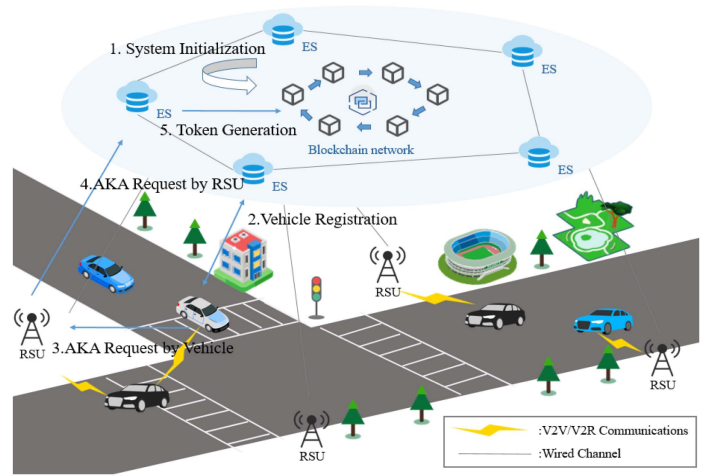


Fig. 1. System model.

addition, the ES provides registration services for vehicles within its service coverage area.

- *RSU*: The RSU provides network access to vehicles and is equipped with an anomaly detection mechanism that evaluates whether a vehicle's communication behavior is malicious based on its historical data.
- *Vehicle*: Vehicles are the primary entities in VANETs. As untrusted parties, their messages are susceptible to attacks; therefore, both the vehicles and their messages involved in the AKA process must be authenticated.

E. Threat Model

In this scheme, consortium blockchain entities are semi-trusted and jointly maintain network security, with attackers potentially being vehicles, RSUs, ESs, or external adversaries. ESs are assumed to honestly follow the protocol but may attempt to infer sensitive information from messages, and we assume compromised ES nodes remain strictly below threshold t to prevent secret reconstruction or token forgery. Passive adversaries can eavesdrop on all public communications without interfering, while active adversaries can launch replay, tampering, and forgery attacks through adaptive interaction. All adversaries are computationally bounded to polynomial time and cannot solve the underlying RLWE or ISIS hard problems within a feasible timeframe, even with quantum-capable resources.

IV. OUR PROPOSED SCHEME

Our proposed scheme consists mainly of four phases: Setup, Registration, Request, and Response. The main interaction logic is illustrated in the figure.

A. System Initialization

- 1) *Blockchain Construction*: The consortium blockchain is jointly managed and operated by multiple entities to establish a distributed shared database with low confirmation latency. In our proposed scheme, RSUs and ESs are the most common semi-trusted entities in the VANETs and are

used as management nodes of the consortium blockchain. Regarding specific management responsibilities, RSUs are primarily responsible for evaluating vehicle communication behavior, while ESs are primarily responsible for providing registration and authentication services for vehicles.

- 2) *Parameter Initialization*: In this phase, the system initiator prepares hash functions, constructs parameters required for the RLWE problem, and algorithms required for the ISIS problem. RSUs use the consortium blockchain in a collaborative manner to complete initialization without a trusted entity. Additionally, the blockchain will create a smart contract for managing registered addresses.

- a) The system initiator generates public parameters q, n, χ, β , where χ is an error distribution satisfying $Pr[\|x\| > \beta : x \leftarrow \chi] \leq \text{negl}(n)$. Select the ring $R_q = \mathbb{Z}_q[x]/(f(x))$, where $f(x) = x^n + 1$ and n is a power of 2. Generate a random ring element $a \leftarrow R_q$. The system initiator sets a threshold t and chooses some secure hash functions, $H_0 : \{0, 1\}^* \rightarrow \mathbb{Z}_q^p$, $H_1 : \{0, 1\}^* \rightarrow \mathbb{Z}_q$, $H_2 : \{0, 1\}^* \times \{0, 1\}^* \times R_q \rightarrow \mathbb{Z}_q^p$, $H_3 : \{0, 1\}^* \times \{0, 1\}^* \times \{0, 1\}^* \rightarrow \mathbb{Z}_q^p$, $H_4 : R_q \times R_q \times \{0, 1\} \times \{0, 1\}^* \times \{0, 1\}^* \rightarrow \{0, 1\}^\lambda$.
- b) Assume the number of authorized ESs is l . For each ES_j , use the $\text{TrapGen}(p, k, \vartheta)$ algorithm to generate a private-public key pair (A_i, T_i) , where A_i is the public matrix and T_i is the private trapdoor, i.e., a basis for $\Lambda_{\vartheta}^\perp(A_i)$. The set of public-private keys for the l ES nodes is $K = \{(A_1, T_1), (A_2, T_2), \dots, (A_l, T_l)\}$.
- c) Each ES generates a polynomial of degree $(t - 1)$, i.e., $f_j(x) = s + a_1x + a_2x^2 + \dots + a_{t-1}x^{t-1}$. Then, each ES computes threshold shares based on secret sharing, i.e., ES_j evaluates for each other ES's identity ID_x , $s_{jx} = f_j(ID_x)$, for distribution to other ESs. Finally, compute $s_x = \sum_{j=1}^l s_{jx}$ and record s_x on the blockchain. After computing s_{jx} , each ES signs the share using its own private key: $\varphi_j = \text{SamplePre}(A_i, T_i, H_0(r, s_{jx}), \sigma)$ where $(r \leftarrow \{0, 1\}^{t_0})$, and sends (r, s_{jx}, φ_j) to other ES nodes. If verification passes, s_{jx} is accepted. All parameters are put on-chain (sent to the blockchain network), and public information will be stored in the genesis block of the consortium blockchain.

B. Registration Phase

- 1) *RSU and ES Registration*: Each ES_j uses a randomly generated number $s_j \in \mathbb{Z}_q^*$ to generate a blockchain address $addr_j$. Each RSU_k uses the $\text{TrapGen}(p, k, \vartheta)$ algorithm to generate a public-private key pair (A_k, T_k) , and T_k is used as the private key to generate the blockchain address $addr_k$. Before the smart contract is deployed, the system initiator writes it into the smart contract. Subsequently, the sender address of the transaction used for authentication should be equal to one of the valid addresses recorded in the blockchain. Furthermore, for each ES_j node, $ID_j = H_1(addr_j)$, which can be regarded as the unique index

Algorithm 1: Vehicle Registration Function.

```

1: Input:  $addr_v$ 
2: Output:  $bool$ 
3: //verify whether the transaction sender is a member of
   the authorized edge server address list
4: if  $msg.sender$  not in  $esAddr$  then
5:   return false
6: else
7:   //vehs is a mapping type
8:    $vehs[msg.sender][addr_v] \leftarrow true$ 
9:   return true //registration successful
10: end if

```

Algorithm 2: Public Key Registration Function.

```

1: Input:  $addr_j, A_v$ 
2: Output:  $bool$ 
3: if  $vehs[addr_j][msg.sender] \neq True$  then
4:   throw "Vehicle not registered under ES"
5: end if
6: //register each public key in the list
7: for  $index \leftarrow 0$  to  $A_v.length - 1$  do
8:    $filter.insert(A_v[index])$ 
9: end for
10: return true //public key registration successful

```

of the ES node, aiming to map the blockchain address to a value in the integer domain for the convenience of calculating Lagrange interpolation coefficients later.

- 2) *Vehicle Registration*: All vehicles need to register with some ES. Use the $\text{TrapGen}(p, k, \vartheta)$ algorithm to generate a public-private key pair (A_v, T_v) , where T_v can be hashed to a fixed-length identifier as the node's blockchain address $addr_v$. Send the corresponding ciphertext $EA_v = \text{Enc}(A_j, addr_v, id_v)$ to ES_j , hiding the real identity of vehicle V_v . Upon receiving the registration request from V_v , ES_j calls the smart contract vehicle registration function (Algorithm 1) by sending a signed transaction $TX_j^1 = \{addr_v\}_{T_i}$ (the transaction signed by the edge server ES_j using its private key T_i), so V_v 's address $addr_v$ will be considered valid.
- 3) *Vehicle Public Key Registration*: The vehicle calls the smart contract public key registration function (Algorithm 2) by sending a signed transaction $TX_v^1 = \{addr_j, A_v\}_{T_v}$ to the blockchain network. The vehicle's public key A_v will be registered on the blockchain, ensuring the trustworthiness of the public key's origin.

C. Request Phase

In this phase, vehicle V_v first initiates an AKA request and sends this request to the roadside unit RSU for processing.

- 1) *Vehicle Side*: The vehicle selects a random secret ring element $r_v \leftarrow \chi$, computes the corresponding public key $X_v = ar_v + 2f_v \bmod q$, where $f_v \leftarrow \chi$, for subsequent key agreement. Then, V_v obtains the current timestamp t_v ,

Algorithm 3: Public Key Verification Function.

```

1: Input:  $h(A_v)$ 
2: Output: bool
3: //check whether  $A_v$  exists in the filter
4: if  $filter.lookup(h(A_v)) == true$  then
5:   return true
6: else
7:   return false
8: end if

```

computes $h_v = H_2(addr_v, t_v, X_v)$, and generates vehicle signature ε_v by invoking the lattice sampling algorithm $SamplePre(A_v, T_v, h_v, \sigma)$ as shown in Section III-A, where A_v has been generated and registered. It then sends the request message $M_v = \{addr_v, t_v, X_v, A_v, \varepsilon_v\}$ to the RSU.

- 2) *RSU Side:* Upon receiving M_v , RSU_k checks the message freshness by determining whether $|t_{cur} - t_v| \leq \Delta T$ holds, where t_{cur} represents the current timestamp and ΔT represents the maximum legal time interval. If not, RSU_k rejects this message and aborts the request. Otherwise, RSU_k selects a random secret ring element $r_{rsu} \leftarrow \chi$, computes the corresponding public key $X_{rsu} = ar_{rsu} + 2f_{rsu} \bmod q$, $f_{rsu} \leftarrow \chi$. Obtains the current timestamp t_k , generates a blockchain transaction $TX_k = \{addr_v, t_v, X_v, A_v, \varepsilon_v, X_{rsu}, t_k\}_{T_k}$ signed with a GPV signature [51], and sends it to the consortium blockchain.

D. Response Phase

In this phase, the AKA request initiated by the vehicle is processed by the ESs, which generate signatures as distributed authentication tokens for this request. If the number of these tokens reaches the threshold, the AKA request is considered valid, meaning that the participants in the AKA session can execute the key agreement process and securely establish a session key.

- 1) *Message Authentication:* Upon receiving TX_k , ES_j first checks the freshness of timestamps t_k and t_v . If either t_k or t_v fails the freshness check, the session is aborted. Second, ES_j is responsible for verifying the validity of the transaction TX_k (by verifying the corresponding GPV signature of the transaction). Subsequently, ES_j calls the public key verification function (Algorithm 3) to check the validity of the vehicle's registered public key A_v . If the function returns false, ES_j rejects this message and aborts the request.

If the return value is true, ES_j computes $h_j = H_2(addr_v, t_v, X_v)$ and checks whether the equation $A_v \varepsilon_v = h_j$ holds. If not, the signature is unverified, ES_j rejects this message and aborts the request. Otherwise, the identity verification process for the request is completed.

- 2) *Token Generation:* Once V_v is considered a legitimate vehicle, ES_j computes $h = H_3(addr_v, addr_k, h(TX_k))$

Algorithm 4: Token Aggregation Function.

```

1: Input:  $e_i, A_i, h(TX_k)$ 
2: Output: bool
3: //check whether the caller is a legitimate edge server
4: if  $msg.sender$  not in  $esAddr$  then
5:   return false
6: else
7:   //retrieve the current aggregation counter for the transaction
8:    $ctr = count[h(TX_k)]$ 
9:   //append the new token to the token list of this transaction
10:   $tokens[h(TX_k)][ctr] \leftarrow (e_i, A_i)$ 
11:  //update the token counter for the transaction
12:   $count[h(TX_k)] \leftarrow ctr + 1$ 
13:  return true //token storage successful
14: end if

```

Algorithm 5: Token Validation Function.

```

1: Input:  $h(TX_k)$ 
2: Output: token[]
3: //get the current number of tokens stored for the transaction
4:  $current \leftarrow count[h(TX_k)]$ 
5: //check if the token count meets the threshold  $t$ 
6: if  $current < t$  then
7:   return NULL //threshold not reached, authentication incomplete
8: else
9:   //return the entire token array associated with the transaction
10:  return  $tokens[h(TX_k)]$ 
11: end if

```

and $h_i = h + s_i$, then uses the lattice sampling algorithm $SamplePre(A_i, T_i, h_i, \sigma)$ to generate a token e_i for V_v satisfying $A_i e_i = h_i$. Finally, ES_j sends the generated share $(e_i, A_i, h(TX_k))$ to the blockchain as input to the token aggregation function (Algorithm 4).

- 3) *Transaction Legitimacy Verification:* Only after the on-chain tokens reach the value t will the corresponding request message be considered authenticated. When verifying the validity of the threshold signature, select a node ES_γ such that $(A_\gamma, T_\gamma) \in K$. ES_γ calls the defined smart contract token validation function (Algorithm 5). If the threshold t is not reached, it returns NULL; otherwise, it returns all tokens. It then computes $C_i \leftarrow SampleRwithBasis(A_\gamma, T_\gamma, A_i, \sigma)$ and $e = \sum_{i=1}^t DC_i e_i \chi_i$, where $e \in \mathbb{Z}_q^p$, the amplification factor $D = (l!)^2$, and the Lagrange interpolation coefficients $\chi_i = \prod_{j=1, j \neq i}^t (-ID_j) / (ID_i - ID_j)$. If $A_\gamma e = Dh$ holds, then the transaction is legitimate, and the vehicle and RSU can proceed with key agreement.

E. Key Agreement

- 1) *Vehicle Side*: In this phase, the vehicle can trust messages received from the ESs and sends related parameters to the RSU, using them to compute the session key. The vehicle randomly samples the error polynomial f'_v from χ and calculates $K_v = X_{rsu}r_v + 2f'_v \bmod q$. Then the scheme utilizes the *Cha* function defined in Definition 2 to calculate $\partial_v = Cha(K_v)$, and it also utilizes the *mod* function defined in Definition 3 to calculate $m_v = Mod_2(K_v, \partial_v)$. The parameter $\tau_v = H_4(X_v, X_{rsu}, m_v)$ is generated for later verification. Finally, the session key is defined as

$$sk_v = H_4(X_v, X_{rsu}, m_v, h(TX_k), \tau_v) \quad (1)$$

where sk_v is regarded as the session key. Finally, V sends $\{\partial_v, \tau_v\}$ to the RSU.

- 2) *RSU Side*: Upon receiving $\{\partial_v, \tau_v\}$, the RSU randomly samples the error polynomial f'_{rsu} from χ and computes $K_{rsu} = X_v r_{rsu} + 2f'_{rsu} \bmod q$. Then RSU utilizes the *mod* function to calculate $m_{rsu} = Mod_2(K_{rsu}, \partial_v)$ and gains $\tau_{rsu} = H_4(X_v, X_{rsu}, m_{rsu})$. if $\tau_v = \tau_{rsu}$ holds, the RSU can obtain the session key

$$sk_{rsu} = H_4(X_v, X_{rsu}, m_{rsu}, h(TX_k), \tau_{rsu}) \quad (2)$$

V. SECURITY ANALYSIS

In this section, we demonstrate the correctness of mutual authentication in the proposed scheme and conduct a formal security analysis to prove that the proposed scheme satisfies session key security.

A. The Correctness of Mutual Authentication

- 1) As mentioned above, if both the message and the corresponding sender have been authenticated, the ES needs to generate a token for the transaction TX_k . AKA participants, including V_i and RSU_k , can use the equation (to be referenced) to check if the requirement is met. This paper utilizes the Lagrange interpolation theorem to prove the correctness of the equation (to be referenced). Verification of equation $A_\gamma e = Dh$ is as follows.

$$\begin{aligned} A_\gamma e &= A_\gamma \sum_{i=1}^t DC_i e_i \chi_i = D \cdot \sum_{i=1}^t A_\gamma C_i e_i \chi_i \\ &= D \cdot \sum_{i=1}^t A_i e_i \chi_i = D \cdot \sum_{i=1}^t (h + s_i) \chi_i \\ &= Dh \sum_{i=1}^t \chi_i + D \sum_{i=1}^t s_i \chi_i \end{aligned} \quad (3)$$

That is, we need to prove $\sum_{i=1}^t \chi_i = 1$ and $\sum_{i=1}^t s_i \chi_i = 0$. By the property of Lagrange interpolation $\chi_i = \prod_{j=1, j \neq i}^t \frac{-ID_j}{ID_i - ID_j}$, we have $s = f(0) = \sum_{i=1}^t s_i \chi_i$. $\sum_{i=1}^t \chi_i = 1$ can be understood as the special case where s_i is always 1, i.e., there exist l points $\{(ID_1, g_1), (ID_2, g_2), \dots, (ID_l, g_l)\}$ from polynomial

$G(x)$, where $g_i = G(ID_i) = 1, i = 1, 2, \dots, l$. Therefore, we can obtain:

$$\begin{aligned} \sum_{i=1}^t \chi_i &= \sum_{i=1}^t 1 \cdot \chi_i = \sum_{i=1}^t g_i \cdot \chi_i \\ &= G(0) = 1 \end{aligned} \quad (4)$$

the equation reduces to:

$$A_\gamma e = Dh \cdot 1 + Dh \cdot 0 = Dh \quad (5)$$

- 2) In this scheme, to verify that the session keys $sk_v = sk_{rsu}$, it suffices to prove that $m_v = m_{rsu}$ for both entities. That is, to apply Lemma 3, we need to prove $\|K_v - K_{rsu}\| \leq (q/4) - 2$. The proof is as follows:

$$\begin{aligned} \|K_v - K_{rsu}\| &= 2(f_{rsu}r_v + f'_v - f_v r_{rsu} - f'_{rsu}) \\ &\leq 2(f_{rsu}r_v + f'_v + f_v r_{rsu} + f'_{rsu}) \\ &\leq 2(\|f_{rsu}\| \cdot \|r_v\| + \|f'_v\| + \|f_v\| \cdot \|r_{rsu}\| + \|f'_{rsu}\|) \\ &\leq 2((\beta\sqrt{n})(\beta\sqrt{n}) + \beta\sqrt{n} + (\beta\sqrt{n})(\beta\sqrt{n}) + \beta\sqrt{n}) \\ &\leq 2(2(\beta\sqrt{n})^2 + 2\beta\sqrt{n}) \\ &\leq 8\beta^2 n \leq (q/4) - 2 \end{aligned} \quad (6)$$

B. Security Model

We define a security framework based on a game-based model, where a series of games are used to illustrate the adversary's behavior and the interaction between the adversary and the challenger. Participants are of three types: vehicle V_v , roadside unit RSU_k , and edge server ES_j . We denote the t -th instance of participant $\Lambda \in \{V_v, RSU_k, ES_j\}$ as Π_Λ^t . In these games, the adversary $\mathcal{A}$ can send queries to multiple oracles and receive corresponding responses from the challenger $\mathcal{C}$. The detailed definitions are as follows.

- *Execute*(Π_V^v, Π_{RSU}^k): Simulates passive attacks (eavesdropping), returns real interaction messages.
- *Send*(Π_Λ^t, m_t): Simulates active attacks (forgery/tampering), allows sending arbitrary messages to an instance.
- *Reveal*(Π_V^v, Π_{RSU}^k): Returns the session key, used to test forward security.
- *Corrupt*(Π_Λ^t): Returns the long-term private key, used to simulate key leakage scenarios.
- *Test*(Π_V^v, Π_{RSU}^k): Core security test, the challenger randomly returns either the real session key or a random value, and the attacker must distinguish.

Definition 6 (Freshness): A session that has not been subjected to a *Reveal* query is fresh.

Definition 7 (Correctness): Two participant instances independently compute the same session key.

Definition 8 (Session Key Semantic Security): The adversary's advantage in winning the *Test* game must be negligible.

C. Proven Security

Theorem 1: If no polynomial-time adversary can break the decisional RLWE hard problem and the ISIS problem, then the proposed AKA protocol satisfies session key semantic security under the defined security model as shown in Section V-B.

Proof: We use the game-hopping technique for the proof. The adversary $\mathcal{A}$ and the challenger $\mathcal{C}$ interact according to predefined rules. Let W_i be the event that the adversary wins in Game_i . The detailed definitions of the games are as follows.

- **Game 1:** In Game 1, $\mathcal{C}$ strictly follows the protocol specifications, generating public parameters q, n, χ, β , where χ is the error distribution satisfying $\Pr[\|x\| > \beta : x \leftarrow \chi] \leq \text{negl}(n)$. Select the ring $R = \mathbb{Z}[x]/(f(x))$, where $f(x) = x^n + 1$, n is a power of 2, and generate a random ring element $a \leftarrow R_q$. Set the threshold t and the total number of ESs l . $\mathcal{C}$ initializes the following empty key-value mappings: $KV_1[\Pi_\Lambda^t]$ is used to store the long-term key, temporary values, and outputs of instance Π_Λ^t when processing specific message msg . $KV_2[\Pi_\Lambda^t]$ is used to store the token of instance Π_Λ^t . $KV_3[\Pi_\Lambda^t]$ is used to store the session key established between instances Π_V^v and Π_{RSU}^k .

Use $\mathcal{A}$ as a subroutine to break the protocol. $\mathcal{A}$ can request oracles defined as follows; here, blockchain-related interactions are simplified to core cryptographic operations.

- 1) $\text{Send}(\Pi_V^v, \Pi_{RSU}^k, \text{reg})$: Before processing any Send queries, $\mathcal{C}$ initializes key-value pairs according to the real protocol. Upon receiving this query, $\mathcal{C}$ first checks if $KV_1[\Pi_V^v]$ is empty. If not, $\mathcal{C}$ returns $KV_1[\Pi_V^v, \Pi_{RSU}^k]$ to $\mathcal{A}$. Otherwise, $\mathcal{C}$ executes the protocol, generating $\text{addr}_v, t_v, A_v, T_v, TX_j^1$ for Π_V^v and t_k, X_k for Π_{RSU}^k . It stores $\{\text{addr}_v, t_v, A_v, T_v, TX_j^1, t_k, X_k\}$ as the value of $KV_1[\Pi_V^v, \Pi_{RSU}^k]$ and sends the tuple to $\mathcal{A}$.
- 2) $\text{Send}(\Pi_V^v, \Pi_{RSU}^k, \{A_v\})$: Upon receiving this query, $\mathcal{C}$ first checks if $KV_1[\Pi_V^v, \Pi_{RSU}^k]$ is empty. If empty, abort the query; otherwise, $\mathcal{C}$ extracts values from $KV_1[\Pi_V^v, \Pi_{RSU}^k]$, selects random ring elements to compute $X_v, X_{rsu}, \varepsilon_v$, generates TX_k , and sends it to $\mathcal{A}$.
- 3) $\text{Send}(\Pi_V^v, \Pi_{RSU}^k, \{TX_k\})$: Upon receiving this query, $\mathcal{C}$ first checks the validity of TX_k using the verification equation (vehicle signature). If verification fails, abort the query; otherwise, $\mathcal{C}$ uses a sampling algorithm to generate tokens $e_i (1 \leq i \leq t)$, stores them in $KV_2[\Pi_V^v, \Pi_{RSU}^k]$, and sends them to $\mathcal{A}$.
- 4) $\text{Send}(\Pi_V^v, \Pi_{RSU}^k, \{TX_k, e_i (1 \leq i \leq t)\})$: Upon receiving this query, $\mathcal{C}$ first checks the validity of the tokens using the verification equation. If unverified, abort the query; otherwise, $\mathcal{C}$ uses a signal function and a fuzzy extractor to compute the session key sk , stores it in $KV_3[\Pi_V^v, \Pi_{RSU}^k]$, sends it to $\mathcal{A}$, and considers the instance no longer fresh.
- 5) $\text{Execute}(\Pi_V^v, \Pi_{RSU}^k)$: Upon receiving this query, $\mathcal{C}$ first checks if $KV_1[\Pi_V^v, \Pi_{RSU}^k]$ and $KV_2[\Pi_V^v, \Pi_{RSU}^k]$ are empty. If either is empty, $\mathcal{C}$ aborts the query; otherwise, $\mathcal{C}$ sends these values to $\mathcal{A}$.

- 6) $\text{Reveal}(\Pi_V^v, \Pi_{RSU}^k)$: Upon receiving this query, $\mathcal{C}$ first checks if $KV_3[\Pi_V^v, \Pi_{RSU}^k]$ is empty. If empty, $\mathcal{C}$ aborts the query; otherwise, $\mathcal{C}$ sends the established session key sk to $\mathcal{A}$.
- 7) $\text{Test}(\Pi_V^v, \Pi_{RSU}^k)$: Upon receiving this query, $\mathcal{C}$ first checks if Π_V^v, Π_{RSU}^k are fresh instances. If not, $\mathcal{C}$ aborts the query; otherwise, $\mathcal{C}$ flips a random coin $b \in \{0, 1\}$. If $b = 1$, $\mathcal{C}$ extracts the value sk from $KV_3[\Pi_V^v, \Pi_{RSU}^k]$ and returns the real session key. If $b = 0$, it selects a random key of the same length as the session key and sends it to $\mathcal{A}$.

The adversary observes the actual protocol interactions and the Test query. This is the real attack environment for the adversary. Clearly, the above process is identical to the original protocol, thus satisfying:

$$\text{Adv}_{\mathcal{A}} = 2 |\Pr(W_1) - 1/2| \quad (7)$$

- **Game 2:** In this game, all interactions are the same as in Game 1, except that all hash functions are simulated as random oracles. Clearly, the adversary's view remains indistinguishable. Allow $\mathcal{A}$ to execute $\text{Send}(\Pi_\Lambda^t, m_t)$ queries by generating, tampering with, and sending requests to Π_V^v, Π_{RSU}^k to launch spoofing attacks. However, these messages embed random elements and timestamps, so $\mathcal{A}$ must avoid any collisions when executing $\text{Send}(\Pi_\Lambda^t, m_t)$ queries. Potential collisions on $\{X_v, X_{rsu}\}$ and $\{K_v, K_{rsu}\}$ need to be considered. According to the birthday paradox, we have:

$$\Pr(W_2) - \Pr(W_1) \leq \frac{q_h^2}{2|\mathcal{P}_{\min}|} + \frac{(q_s + q_e)^2}{2|S_{\chi\beta}|} \quad (8)$$

- **Game 3:** In this game, all interactions are the same as in Game 2, except for resisting existential unforgeability under adaptive chosen-message attacks (UF-CMA). We prove that the authentication part of the protocol is unforgeable in the random oracle model. Assume there exists a polynomial-time adversary $\mathcal{A}$ that can forge a valid token with non-negligible probability ϵ , breaking the scheme's security. We use $\mathcal{A}$ to construct a simulator $\mathcal{F}$ to solve the ISIS problem.

- 1) $\mathcal{F}$ receives an ISIS challenge instance (A^*, y) , where $A^* \in \mathbb{Z}_q^{p \times k}, y \in \mathbb{Z}_q^p$. It attempts to find a short vector e^* such that $A^* e^* = y$ and $\|e^*\| \leq \beta$. $\mathcal{F}$ sets the public key of a specific node ES_{j^*} to $A_{j^*} = A^*$. For all other $ES_j (j \neq j^*)$, $\mathcal{F}$ normally executes the $\text{TrapGen}()$ algorithm to generate their public-private key pairs (A_i, T_i) , so $\mathcal{F}$ knows the trapdoors for these ESs. $\mathcal{F}$ interacts with adversary $\mathcal{A}$ on behalf of all honest parties (vehicles, RSUs, and ESs except j^*).
- 2) $\mathcal{F}$ runs $\mathcal{A}$ (as a subroutine) for the first time. $\mathcal{F}$ possesses the random oracle H_3 list L_{H_3} . Suppose in this run, $\mathcal{A}$ eventually outputs a forged signature e for message m^* , and it has queried H_3 once, obtaining response $h_1 = H_3(m^*)$, with internal state s_1 at that moment. $\mathcal{F}$ performs a second run, rolling back from s_1 , but provides a different, random, and independent hash

value h_2 for m^* . According to the General Forking Lemma, if $\mathcal{A}$ succeeds in forging with probability ϵ in the first run, then the probability that it outputs another valid forgery e' for the same message m^* in the second run is at least $\epsilon^2/\text{poly}(n)$.

- 3) From the forking lemma, we obtain $A^*(e - e') = D(h_1 - h_2)$, so $e - e' = A^{*-1}D(h_1 - h_2)$. According to the properties of the lattice basis sampling algorithm, the lengths of the original signature vectors e and e' are strictly bounded, satisfying $\|e\| \leq t \cdot (l!)^3 \cdot O(k^{2.5}) \cdot \omega(\log k) = \beta_0$. Therefore, the upper bound of the norm of the difference between the two signature vectors is $\|e - e'\| \leq \|e\| + \|e'\| \leq 2\beta_0 = 2t \cdot (l!)^3 \cdot O(k^{2.5}) \cdot \omega(\log k) = \beta$. We denote $e^* = e - e'$ as the candidate solution to the ISIS problem. Since $h_1 \neq h_2$ and D is non-zero, this vector is necessarily non-zero and satisfies $A^*(e - e') = D(h_1 - h_2)$ and the norm constraint $\|e - e'\| \leq \beta$, so e^* is a solution to the ISIS instance (A^*, y) . In the random oracle model, q_h is the maximum number of hash queries $\mathcal{A}$ can make, considering the hash collision probability $\epsilon \leq q_h/(1 - 1/q_h)^{q_h} \cdot \epsilon'$. By the forking lemma, the challenger's probability of simulating a solution to the ISIS hard problem is $\epsilon_{ISIS} \geq \epsilon \cdot (1 - 2^{-\omega(\log n)})$. Since ϵ_S is negligible, contradicting the assumption, the probability ϵ of forging a signature must be negligible.

Thus, it satisfies:

$$\Pr(W_3) - \Pr(W_2) \leq \epsilon \leq \frac{\epsilon_{ISIS} \cdot q_h}{\left(1 - \frac{1}{q_h}\right)^{q_h} \cdot (1 - 2^{-\omega(\log n)})} \quad (9)$$

- **Game 4.** Game 4 is the same as Game 3, except that the session key uses the concept of RLWE, so the key establishment relies on the hash oracle. The creation of the session key $sk_{rsu} = H_4(X_v, X_{rsu}, m_{rsu}, h(TX_k), \tau_{rsu})$ depends on random $\{r_v, f_v, r_{rsu}, f_{rsu}, f'_v, f'_{rsu}\}$, which are only chosen by Π_V^v , Π_{RSU}^k during key agreement. Furthermore, Π_V^v and Π_{RSU}^k do not store $\{r_v, f_v, r_{rsu}, f_{rsu}, f'_v, f'_{rsu}\}$ after use. To compute sk , $\mathcal{A}$ must compute $\{K_v, K_{rsu}\}$ from the public $\{X_v, X_{rsu}\}$, which essentially requires solving the RLWE problem, i.e., recovering r_v from $X_v = ar_v + 2f_v$ or r_{rsu} from $X_{rsu} = ar_{rsu} + 2f_{rsu}$. Therefore, we have:

$$\Pr(W_4) - \Pr(W_3) \leq (q_s + q_e) \text{Adv}_A^{\text{RLWE}} \quad (10)$$

By combining the several games, we obtain the equation:

$$\begin{aligned} \text{Adv}_A &= 2|\Pr(W_1) - 1/2| \\ &= 2|\Pr[W_1] - \Pr[W_2] + \Pr[W_2] \\ &\quad - \Pr[W_3] + \Pr[W_3] - \Pr[W_4]| \\ &\leq 2(|\Pr[W_2] - \Pr[W_1]| + |\Pr[W_3] - \Pr[W_2]| \\ &\quad + |\Pr[W_4] - \Pr[W_3]|) \\ &\leq \frac{q_h^2}{2|\mathcal{P}_{\min}|} + \frac{(q_s + q_e)^2}{2|S_{\chi\beta}|} + \frac{\epsilon_{ISIS} \cdot q_h}{\left(1 - \frac{1}{q_h}\right)^{q_h} \cdot (1 - 2^{-\omega(\log n)})} \end{aligned}$$

$$+ (q_s + q_e) \cdot \text{Adv}_A^{\text{RLWE}} \quad (11)$$

D. Security and Privacy Requirements

- 1) **Mutual Authentication:** In the response phase, ESs verify the validity of TX_k, A_v, ε_v via signature verification. To also prevent internal attacks from vehicles, ESs authenticate V_v by analyzing its historical communication data or logs with help from nearby RSUs. After successful vehicle authentication, the ES generates an authentication token for TX_k . When TX_k obtains signature authentication tokens from t nodes from various ESs, both V_v and RSU_k can authenticate TX_k using the verification equation. This authentication process ensures the legitimacy and integrity of AKA messages, as proven in the theorem. Therefore, the protocol achieves authentication of transmitted messages and their corresponding senders.
- 2) **Confidentiality:** The protocol generates session keys $h(X_v, X_{rsu}, m_{rsu}, h(TX_k), \tau_{rsu})$ and $h(X_v, X_{rsu}, m_v, h(TX_k), \tau_v)$, which can only be computed by V_v and RSU_k . Since $\{X_v, X_{rsu}, K_v, K_{rsu}\}$ are necessary for generating the session key, and the private keys r_v and r_{rsu} required to compute these values are held only by the users, no entity other than V_v and RSU_k can compute the session key. Therefore, the protocol ensures confidentiality.
- 3) **Resistance to Replay Attacks:** The protocol defends against such attacks by embedding timestamps t_v and t_k in the transaction TX_k : the receiver first validates the timestamp freshness, discarding expired messages, and then verifies message integrity via hashing to prevent tampering. These two verification steps collectively provide effective resistance against replay attacks.
- 4) **Resistance to Man-in-the-Middle Attacks:** The protocol prevents MitM attacks with signed transactions safeguarding critical parameters like addresses, keys, and timestamps. Any tampering causes authentication failure, ensuring secure session key generation.
- 5) **Resistance to DDoS Attacks:** A DDoS attack floods a server with fake requests, disrupting normal services. The protocol employs a consortium blockchain-based distributed authentication network maintained by multiple servers, avoiding single points of failure. Moreover, timestamps enable message freshness verification, filtering malicious requests early and reducing server load, thereby mitigating DDoS attacks.
- 6) **Resistance to Collusion Attacks:** A collusion attack involves multiple malicious nodes cooperating to deceive the system. In this protocol, all ESs have a fully peer-to-peer and independent relationship. Since these ESs are typically maintained by different infrastructure providers or management authorities, the difficulty for an attacker to simultaneously compromise multiple independently managed servers is much higher than attacking ordinary vehicle nodes. Meanwhile, according to the definition of the polynomial, an attacker must control at least t ES nodes to reconstruct the secret and forge authenticated

TABLE II
SECURITY AND PRIVACY COMPARISONS

	R1	R2	R3	R4	R5	R6	R7	R8	R9	R10	R11
Khalid et al.'s scheme [33]	✓	✓	✓	✓	✓	×	✓	✓	✓	✓	⊗
Liu et al.'s scheme [24]	✓	✓	✓	✓	✓	×	✓	✓	✓	✓	⊗
Cui et al.'s scheme [25]	✓	✓	✓	✓	✓	×	✓	✓	×	✓	⊗
Wei et al.'s scheme [23]	✓	✓	✓	✓	×	×	✓	✓	✓	×	⊙
Zhong et al.'s scheme [40]	✓	✓	✓	✓	×	×	✓	✓	✓	✓	⊗
Kumar et al.'s scheme [41]	✓	✓	✓	✓	×	×	✓	✓	✓	×	⊗
Our proposed scheme	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	⊙

✓: The requirement is satisfied ⊗: Require fully-trusted environment
×: The requirement is not satisfied ⊙: Require semi-trusted environment

AKA messages. In the actual IoV environment, as long as the value of t is set reasonably, such a large-scale node collusion is considered technically and economically infeasible. Even if some ESs are compromised, the system remains secure.

- 7) *Resistance to Key Exposure Attacks*: In key exposure attacks, attackers obtain the private key through theft, side-channel analysis or system vulnerabilities. The protocol is capable of resisting such risks, ESs use private keys solely to generate initial authentication tokens, which do not contribute to sessionkey derivation. Hence, even if an ES key is leaked, previously established session keys remain secure, preserving confidentiality and integrity.
- 8) *Forward and Backward Secrecy*: Forward and backward secrecy ensures that compromising a current session key does not expose past or future communications. Since vehicles and RSUs use randomly generated, onetime private keys for each session, the resulting session keys are independent across sessions, thereby achieving both forward and backward secrecy.
- 9) *Perfect Forward Secrecy*: In our proposed AKA scheme, the long-term session key possessed by the vehicle is only used to generate messages that can be verified by the receiver. In other words, the long-term session key is separated from the key agreement process.
- 10) *Conditional Privacy Preservation*: Conditional privacy protection ensures that a pseudoidentity can be linked to the real identity only by authorized entities. The vehicle's identity is stored in ciphertext, so attackers cannot decrypt intercepted messages. When malicious activity occurs, the ES can decrypt the data to reveal the vehicle's true identity, thereby ensuring conditional privacy.
- 11) *Required Trust Level of Servers*: The trust level required for an entity refers to its needed credibility during authentication. Our scheme employs multiple alliance blockchain entities for distributed authentication of AKA requests. This ensures continued security even if one entity goes offline, thus only semi-trusted entities are required. In contrast, schemes like those in [24], [25], [33] rely on a single authenticating entity, which necessitates full trust in that entity.

E. Security and Privacy Comparisons

In this section, we conduct a comparative analysis of our proposed scheme and several related schemes, including the ones by Khalid et al. [33], Liu et al. [24], Cui et al. [25], Wei et al. [23], Zhong et al. [40], and Kumar et al. [41], to meet security and privacy requirements. Let R1, R2, R3, R4, R5, R6, R7, R8, R9, R10, and R11 represent mutual authentication, confidentiality, resistance to replay attacks, resistance to man-in-the-middle attacks, resistance to DDoS attacks, resistance to collusion attacks, resistance to key exposure attacks, forward and backward secrecy, perfect forward secrecy, conditional privacy protection, and the required trust level of the server. Table II provides a comprehensive comparison of security and privacy aspects, indicating that the proposed scheme better meets the security and privacy requirements than the other schemes.

VI. PERFORMANCE ANALYSIS

In this section, we analyze the computational and communication overhead of the proposed scheme. Specifically, the testbed is a PC equipped with Ubuntu 22.04.3, an i7-12700 CPU, and 32 GB RAM. Communication performance in the VANETs environment is measured using the Veins framework [52], which is based on the network simulation platform OMNeT++ [53] and the traffic simulation software SUMO [54], and use the NFlib cryptographic library [55] to measure the execution time of cryptographic operations. Additionally, blockchain-related performance is measured using the Ganache [56] and Truffle frameworks, which serve as the development environment, testing framework, and asset pipeline for Ethereum. Comparisons are made with six schemes proposed by Khalid et al. [33], Liu et al. [24], Cui et al. [25], Wei et al. [23], Zhong et al. [40], and Kumar et al. [41].

Our scheme performs computations in the ring $R_q = \mathbb{Z}_q[x]/(x^n + 1)$ with dimension $n = 1024$ and $q = 2147483647$. We use a discrete Gaussian distribution with standard deviation $\sigma = 3.192$. Any ring element in R_q uses 31774 bits. Meanwhile, we use the SHAKE256 algorithm to generate a 256-bit hash output. Based on the estimates in [23], we set the system parameters $p = 6$, $k = 256$, $\vartheta = 16777215$.

TABLE III
EXECUTION TIME OF SEVERAL CRYPTOGRAPHIC OPERATIONS

Operation	Description	Run time (ms)
T_{add}	Execution time of the addition of two polynomial elements in R_q	0.0002
T_{mul}	Execution time of the multiplication of two polynomial elements in R_q	0.0022
T_{mul-m}	Execution time of the multiplication of an element in $Z_q^{p \times k}$ and an element in Z_q^k	0.0075
T_{cha}	Execution time of $Cha(*)$ in R_q	0.022
T_{mod}	Execution time of $Mod_2(*)$ in R_q	0.035
T_{h0}	Hash function $H_0: \{0, 1\}^* \rightarrow Z_q^n$	0.0095
T_{h1}	Hash function $H_1: \{0, 1\}^* \rightarrow Z_q$	0.0005
T_{h2}	Hash function $H_2: \{0, 1\}^* \times \{0, 1\}^* \times R_q \rightarrow Z_q^p$	0.0035
T_{h3}	Hash function $H_3: \{0, 1\}^* \rightarrow Z_q^p$	0.0006
T_{h4}	Hash function $H_4: \{0, 1\}^* \rightarrow \{0, 1\}^\lambda$	0.0054

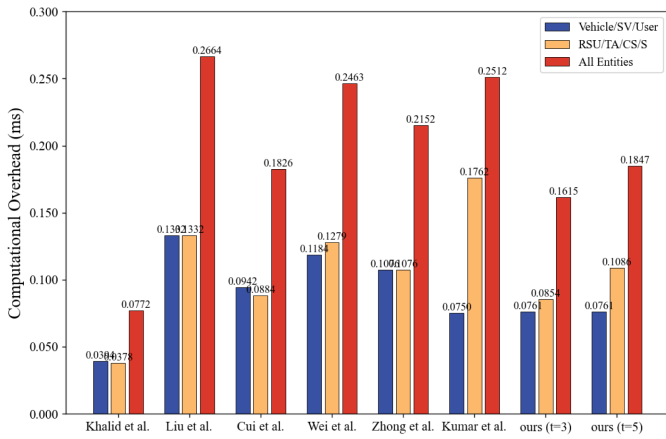


Fig. 2. Computational overhead comparison.

A. Computational Overhead

In this subsection, to facilitate the comparison of the aforementioned schemes, we employ a lattice-based cryptography library named NTLlib to measure all operations. The average execution time of each operation is obtained by running 1000 trials, where the input is processed in a single batch. The detailed execution times of each operation are recorded in Table III.

We primarily consider the time cost of the proposed scheme during the mutual AKA process and compare it with the schemes of Khalid et al. [33], Liu et al. [24], Cui et al. [25], Wei et al. [23], Zhong et al. [40], and Kumar et al. [41] in Table IV and Fig. 2. Specifically, Fig. 2 shows the computational cost for the vehicle and other entities, as well as the total computational cost for all entities. Note that we do not consider the time spent on pre-computations that can be performed in advance to optimize the AKA execution time of the proposed and related schemes.

The comparison of computational costs is shown in Fig. 2. From Fig. 2, the computational cost of our proposed scheme is comparable to that of the schemes by Cui et al.'s scheme [25] and Wei et al.'s scheme [23], and is significantly better than those of Liu et al.'s scheme [24], Zhong et al.'s scheme [40],

and Kumar et al.'s scheme [41]. The results indicate that our scheme occupies fewer computing resources than most existing methods and achieves an excellent balance between efficiency and security.

B. Communication Overhead

In this subsection, to compare the proposed scheme with related works Khalid et al. [33], Liu et al. [24], Cui et al. [25], Wei et al. [23], Zhong et al. [40], and Kumar et al. [41] in terms of communication overhead, we set the output size of the address and hash function to 256 bits, the length of the timestamp to 64 bits, and the length of each polynomial element to $n * \lceil \log_2(q) \rceil = 31774$ bits. We analyze the communication overhead of our proposed scheme by comparing it with six schemes.

In Khalid et al.'s scheme [33], SV_i needs to send $\{T_{SV_j}, M_1, M_2, TS_{sv}\}$ and $\{M_6, TS_{SV'}\}$ to CS. Therefore, the communication overhead for SV_i is $|T_{SV_j}| + |M_2| + |M_6| + |TS_{SV'}| + 2|R_q| = 8036$ bytes. CS needs to send $\{M_3, M_4, M_5, TS_{cs}\}$ to SV_i , so the communication overhead for CS is $|M_3| + |M_4| + |M_5| + |R_q| = 4064$ bytes. Therefore, the total communication overhead is 12100 bytes.

In Liu et al.'s scheme [24], the vehicle needs to send $\{a_v, p_v, x_v, t_0, \sigma_v\}$ and $\{w_v, k_v\}$ to the RSU. Therefore, the vehicle's communication overhead is $|t_0| + |\sigma_v| + |w_v| + |k_v| + 3|R_q| = 20052$ bytes. The RSU needs to send $\{a_{rsu}, p_{rsu}, x_{rsu}, t_1, \sigma_{rsu}, w_{rsu}, k_{rsu}\}$, so the RSU's communication overhead is $|t_1| + |\sigma_{rsu}| + |w_{rsu}| + |k_{rsu}| + 3|R_q| = 20052$ bytes. Therefore, the total communication overhead is 40104 bytes.

In Cui et al.'s scheme [25], the vehicle needs to send $\{PID_i, t'_i, x_i, c, M_i\}$ to DT, so the vehicle's communication overhead is $|PID_i| + |t'_i| + |M_i| + 2|R_q| = 8004$ bytes. DT needs to send $\{DID_j, t_j, PID_i, t'_i, x_i, x_j, c, M_i, M_j\}$ to TA and send $\{\sigma_k, t'_k, t'_j, x_j, w_j\}$ to the vehicle. Therefore, DT's communication overhead is $|DID_j| + |PID_i| + 4|t| + 2|M| + |\sigma_k| + |w_j| + 4|R_q| = 21176$ bytes. TA needs to send $\{\sigma_k, r_k, t'_k\}$ to DT, so TA's communication overhead is $|\sigma_k| + |r_k| + |t'_k| = 68$ bytes. Therefore, the total communication overhead is 29248 bytes.

In Wei et al.'s scheme [23], communication entity A needs to send $\{C_A, \partial_A, h_A\}$ and $\{T_A, \partial_B\}$, where the size of matrix C_A is $k * k * \lceil \log_2(\vartheta) \rceil = 196608$ bytes. Therefore, A's communication overhead is $|C_A| + |\partial_A| + |h_A| + |T_A| + |\partial_B| = 196736$ bytes. Communication entity B needs to send $\{C_B, h_B\}$, so B's overhead is $|C_B| + |h_B| = 196640$ bytes. Therefore, the total communication overhead is 393376 bytes.

In Zhong et al.'s scheme [40], communication entity A needs to send $\{U_A, \delta_A, TS_A, EID_A\}$ and $\{KC_A\}$, where the size of U_A is $k \times n \times \lceil \log_2(q) \rceil = 7936$ bytes. Therefore, the communication overhead for A is $|U_A| + |\delta_A| + |TS_A| + |EID_A| + |KC_A| = 8136$ bytes. Communication entity B needs to send $\{U_B, \delta_B, TS_B, KC_B\}$, so the overhead for B is $|U_B| + |\delta_B| + |TS_B| + |KC_B| = 8104$ bytes. Consequently, the total communication overhead amounts to 16240 bytes.

TABLE IV
COMPUTATIONAL COST COMPARISON

Scheme	V's Time Cost(ms)	RSU/TA/CS's Time Cost(ms)	Total Cost(ms)
Khalid et al.'s scheme [33]	$2T_{add} + 3T_{mul} + 6T_{h4} = 0.0394$	$7T_{h4} = 0.0378$	0.0772
Liu et al.'s scheme [24]	$5T_{add} + 6T_{mul} + T_{cha} + 2T_{mod} + 5T_{h4} = 0.1332$	$5T_{add} + 6T_{mul} + T_{cha} + 2T_{mod} + 5T_{h4} = 0.1332$	0.2664
Cui et al.'s scheme [25]	$2T_{add} + 2T_{mul} + 6T_{h4} + T_{cha} + T_{mod} = 0.0942$	$2T_{add} + 2T_{mul} + 9T_{h4} + T_{mod} = 0.0884$	0.1826
Wei et al.'s scheme [23]	$2T_{add} + 2T_{mul} + 4T_{h4} + T_{cha} + 2T_{mod} = 0.1184$	$2T_{add} + 2T_{mul} + T_{h0} + 4T_{h4} + T_{cha} + 2T_{mod} = 0.1279$	0.2463
Zhong et al.'s scheme [40]	$2T_{add} + 2T_{mul} + 2T_{h4} + T_{cha} + 2T_{mod} = 0.1076$	$2T_{add} + 2T_{mul} + 2T_{h4} + T_{cha} + 2T_{mod} = 0.1076$	0.2152
Kumar et al.'s scheme [41]	$3T_{add} + 3T_{mul} + 2T_{h4} + T_{cha} + T_{mod} = 0.075$	$7T_{add} + 7T_{mul} + 6T_{h4} + T_{cha} + 3T_{mod} = 0.1762$	0.2512
Ours	$2T_{add} + 2T_{mul} + T_{h2} + 2T_{h4} + T_{cha} + T_{mod} = 0.0761$	$2T_{add} + 2T_{mul} + (T_{mul-m} + T_{h2} + T_{h3}) \cdot t + 2T_{h4} + T_{mod} = 0.0116t + 0.0506$	$0.0116t + 0.1267$

In Kumar et al.'s scheme [41], the communication entity on the MTU side needs to send $\{PID_m, p_m, q_m, \alpha_m, \sigma_m, t_1\}$, where the size of the signature is $n \times \lceil \log_2(q) \rceil + L_H = 4000$ bytes. Therefore, the communication overhead for the MTU is $|PID_m| + |p_m| + |q_m| + |\alpha_m| + |\sigma_m| + |t_1| = 8200$ bytes. The communication entity on the RTU side needs to send $\{ID_r, p_r, q_r, \alpha_r, \sigma_r, t_3\}$, so the overhead for the RTU is $|ID_r| + |p_r| + |q_r| + |\alpha_r| + |\sigma_r| + |t_3| = 8200$ bytes. Consequently, the total communication overhead amounts to 16400 bytes.

In our proposed scheme, the vehicle needs to send $\{addr_v, t_v, X_v, A_v, \varepsilon_v\}$ and $\{\partial_v, \tau_v\}$, where the size of matrix A_v is $p * k * \lceil \log_2(\vartheta) \rceil = 4608$ bytes. Therefore, the vehicle's communication overhead is $|addr_v| + |t_v| + |R_q| + |A_v| + |\varepsilon_v| + |\partial_v| + |\tau_v| = 13512$ bytes. The RSU needs to send the transaction message $\{addr_v, t_v, X_v, A_v, \varepsilon_v, X_{rsu}, t_k\}$, so the overhead is $|addr_v| + 2|t| + 2|R_q| + |A_v| + |\varepsilon_v| = 13360$ bytes. Additionally, the vehicle/RSU selects any ES node to compute the cost of t tokens, so it needs to extract tokens via Algorithm 5. Other ESs need to send the generated $t-1$ tokens $\{h(TX_k), e_i\}$ along with their respective signatures to it, i.e., $(h(TX_k) + e_i) \cdot (t-1) = (t-1) * 800$ bytes. Therefore, the total communication overhead is 30072 bytes.

Based on the above analysis, the comparison of communication overhead is shown in Fig. 3. From the figure, it can be seen that the communication cost of our proposed scheme is comparable to other comparative schemes. The reason for the higher communication cost compared to other post-quantum protocols [25], [33] is the linear increase in token generation overhead, which can be mitigated by choosing an appropriate threshold to trade off performance, thus improving security.

C. Network Simulation Analysis

To analyze the practical performance of our proposed scheme, this subsection constructs a scenario based on a real-world area of 2.5 square kilometers, which involves 100 vehicles communicating with two RSUs via the IEEE 802.11p protocol. The simulation parameters are shown in Table V. Each vehicle sends an AKA authentication request message to the RSUs at random

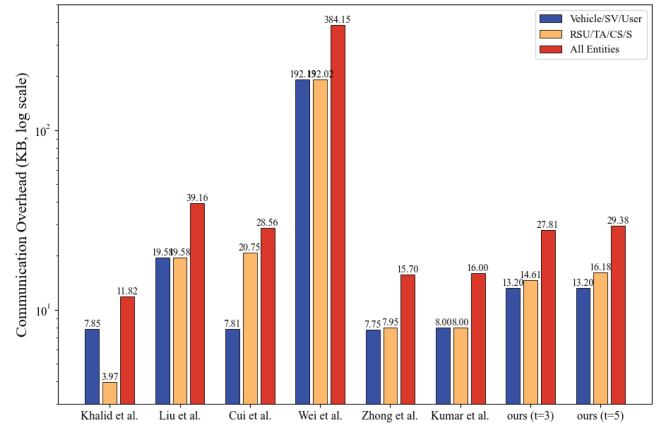


Fig. 3. Communication overhead comparison.

TABLE V
SIMULATION PARAMETERS

Parameter	Value
Simulation area	$2500 \times 2500 (m^2)$
Path loss model	Two-Ray interference
Obstacle shadowing model	Simple obstacle shadowing
Transmission power	50 mW
Data transmission rate	6 Mbps
Sensitivity	-89 dBm
Thermal noise	-110 dBm
Used channel	CCH
Number of vehicles	10 ~ 100
Acceleration speed of vehicle	20 m/s ²
Simulation time	200 s

intervals slightly longer than one second, thereby simulating frequent, yet random, request behavior. Upon receiving a request, the RSU responds via unicast communication.

We tested all schemes with vehicle densities that vary and analyzed changes in transmission delay and packet loss rate during the entire authentication and key agreement process. In this example, the vehicle speed is set to 20 m/s, and the vehicle density (i.e., the number of vehicles) increases from 10 to 100. As shown in Fig. 4, the transmission delay required to complete message authentication and reception increases with increasing

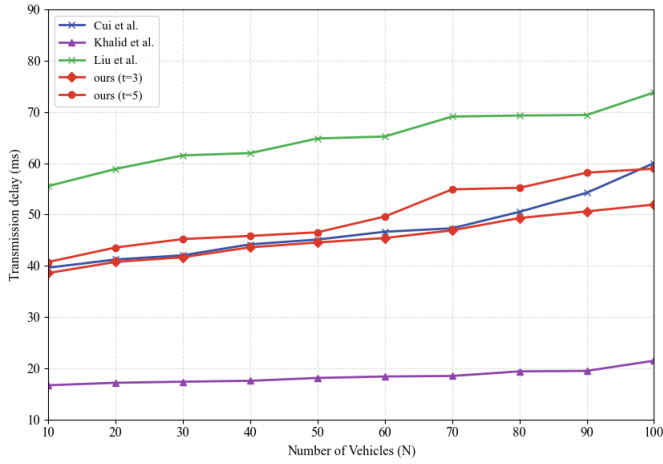


Fig. 4. Impact of vehicle density on transmission delay for each scheme.

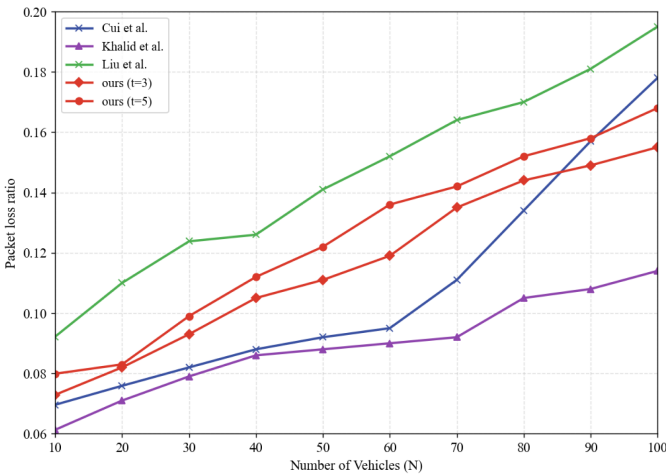


Fig. 5. Impact of vehicle density on packet loss rate for each scheme.

vehicle density for all schemes. The simulation results in Fig. 5 indicate that the packet loss rate during the vehicle authentication process also increases as the density of the vehicle increases. From Figs. 4 and 5, it can be observed that with a default threshold of 5, the transmission delay of our proposed scheme is higher than that of the scheme by Khalid et al.'s scheme [33], but slightly lower than other schemes. Meanwhile, its packet loss rate is lower than that of the scheme by Liu et al.'s scheme [24], but higher than the remaining schemes. When the density of the vehicle reaches 100, the maximum transmission delay of the proposed scheme does not exceed 60 ms, and the maximum packet loss rate does not exceed 16%.

The simulation results demonstrate that our solution achieves a favorable balance between computational efficiency and communication overhead. Notably, the low computational burden on vehicles ensures applicability to resource-constrained on-board units, while the communication cost remains within a controllable range under typical wireless network conditions. From an application perspective, the proposed scheme is effective in intelligent transportation systems, real-time traffic safety information dissemination, and autonomous driving coordination. Furthermore, the decentralized architecture leveraging

TABLE VI
GAS COSTS FOR BLOCKCHAIN OPERATIONS

Operations	Gas
Contract Deployment	1819581
Filter Initiation	$31462.62 \cdot i + 4811.88 \cdot b + 166835.01$
Vehicle Registration	46781
Public Key Registration	134778
Token Registration	115868
Public Key Revocation	40725

edge servers and consortium blockchain enhances scalability and robustness, which is particularly beneficial in large-scale and dynamic vehicular environments. Importantly, existing vehicular communication frameworks standardized by 3GPP and promoted by ITU adopt a layered architecture for V2V, V2I, and V2N communications [57], [58], and ITS security standards such as ETSI TS 102 940 [59] position security mechanisms above the communication stack. Consequently, our scheme can be seamlessly integrated at the security layer without modifying underlying protocols, ensuring compatibility with current systems.

D. Blockchain-Related Cost Analysis

Since our proposed AKA scheme relies on the blockchain, we used Ganache to build a consortium blockchain network and the Truffle framework to measure the actual performance. Our evaluation includes two types: gas cost and processing latency. Gas cost refers to the gas used for state-changing blockchain operations, and processing latency refers to the distance between the request initiation time and the response time.

For the gas cost, we obtained statistics from transaction receipts and the overall magnitude is stable. The initialization cost is linearly related to the capacity parameters and can be approximated by a binary linear equation: $Init \approx 31462.62 \cdot i + 4811.88 \cdot b + 166835.01$ (where i is the number of buckets, b is the number of slots per bucket). This phenomenon is consistent with the mechanism of the contract allocating storage for dynamic arrays row by row per bucket during the initialization phase. Meanwhile, the gas for various registration/revocation operations remains basically unchanged under different capacity configurations, demonstrating good predictability and scalability, as shown in Table VI.

For processing latency, we set up an experiment by simulating 100 vehicles sending different transactions to trigger the execution of state-changing blockchain operations. The results are shown in Fig. 6. From Fig. 6, it can be seen that due to the task scheduling mechanism of the operating system and the diversity of transaction processing in the blockchain network, the processing latency has certain fluctuations. Furthermore, the average processing latencies for vehicle registration, public key registration, token registration, and public key revocation operations are 413.41 ms, 178.3 ms, 290.99 ms, and 201.31 ms, respectively.

According to the experiments, the processing latency in our proposed scheme meets the specified maximum latency standard (500 ms) for V2X communication in high-latency,

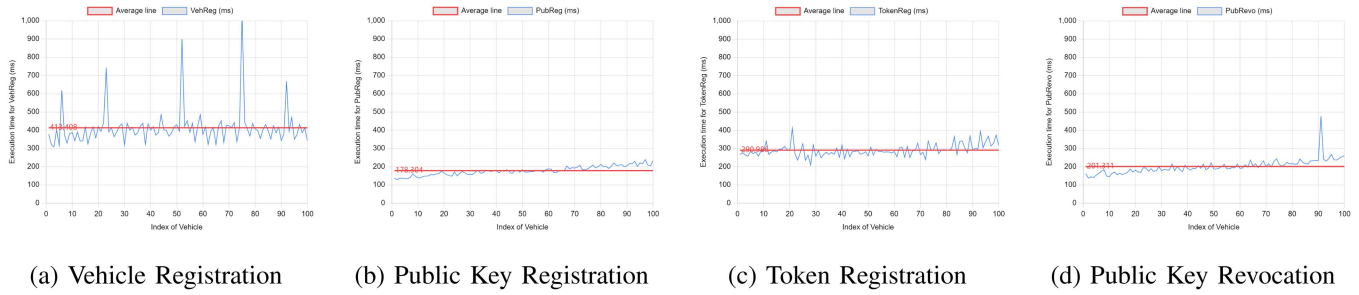


Fig. 6. Performance comparison of different operations in the proposed scheme.

low-frequency category applications, as outlined in T/CSAE 53-2020 [60]. Furthermore, considering the relatively low frequency of AKA usage in the real world, the gas cost in our scheme is considered reasonable. Therefore, it can be concluded that our scheme can meet the requirements of most applications in VANETs.

VII. CONCLUSION

We proposed a decentralized AKA scheme based on the Ring Learning with Errors (RLWE) and Inhomogeneous Small Integer Solution (ISIS) security assumptions for consortium blockchain-enabled VANETs, addressing the security challenges faced by Vehicular Ad-hoc Networks in the quantum era. By introducing a consortium blockchain and a threshold-based voting mechanism, our scheme successfully eliminates the reliance on a single trusted authority, effectively mitigating the single point of failure problem, while leveraging the inherent advantages of lattice-based cryptography to achieve resistance against quantum attacks. Furthermore, our scheme includes rigorous proofs of correctness, security, and compliance with fundamental security and privacy requirements, demonstrating its resilience against various known attacks. Finally, the results of the performance analysis confirm the feasibility of our proposed scheme in terms of computational and communication overhead, featuring acceptable computational cost and reasonable, albeit relatively larger, communication cost.

In future work, we plan to investigate more efficient lattice-based AKA schemes to further reduce communication overhead while preserving strong post-quantum security, for example by optimizing parameter selection and exploring signature aggregation techniques for large-scale vehicular networks. Meanwhile, dynamic threshold adjustment strategies will be studied to better adapt to the highly dynamic topology of VANETs, where factors such as vehicle density, mobility patterns, and link reliability vary over time, requiring a balance between security, efficiency, and fault tolerance. In addition, we will explore the integration of the proposed scheme with practical C-V2X systems, including compatibility with existing standards and deployment considerations, to further validate its applicability in real-world intelligent transportation scenarios.

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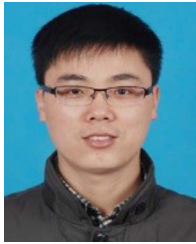


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